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1.1 Prerequisites

Lemma 1.1.1 (Smooth Urysohn's lemma, in mathlib I think)**.** *Let M be a $\mathcal{C}^k$-manifold and A and B two disjoint closed sets in M. Then there is a C^k-function $f\colon M \to [0,1]$ such that f is 0 on A and 1 on B.*

Remark 1.1.2. Again, this requires a definition of manifold that makes it metrizable.

1.2 Basics about collar neighbourhoods

Definition 1.2.1. A *topological manifold* is a second-countable Hausdorff space that is locally Euclidean.

Remark 1.2.2. Apparently, there are some subtleties with collars of non-metrizable manifolds but I assume we don't care. (At least with the above definition our manifolds are always metrizable.)

Definition 1.2.3. Let M and N be two C^k-manifolds. Let $f\colon M \to N$.

(i) f is a C^k-diffeomorphism if it is bijective and both f and its inverse functions are k-times continuously differentiable.

(ii) f is a C^k-embedding if it is a C^k-diffeomorphism onto its image.

Definition 1.2.4 (Due to [Con71])**.** Let M be a $\mathcal{C}^k$-manifold (with boundary) and let $B \subseteq M$ be closed.

(i) B is *locally collared* if it is covered by sets $U \subseteq B$ which are open in B such that there are $\mathcal{C}^k$-embeddings $\varphi_U\colon \overline{U} \times [0,1] \to M$ with

  a) $\varphi_U^{-1}(B) = \overline{U} \times \{0\}$.

  b) For all $x \in \overline{U}$, $\varphi_U(x,0) = x$.

  c) $\varphi_U(\overline{U} \times [0,1])$ is closed.

  d) $\varphi_U(U \times [0,1)$ is open.

(ii) B is *collared* if B already fulfils the conditions given above. In this case we call the image of $B \times [0,1)$ under the corresponding map $\varphi_B\colon B \times [0,1] \to M$ a *collar neighbourhood* of B.

Remark 1.2.5. This definition is a lot more restrictive than other definition which don't consider the closure or the closed interval. I think if this one doesn't work, I can just take the other one.

Remark 1.2.6. For $k = 0$ we can even extend this definition to topological spaces instead of just defining it for topological manifolds.

Theorem 1.2.7 (Due to [Con71])**.** *Let X be a Hausdorff space and $B \subseteq X$ be a closed subset. If B is locally collared in X, then it is collared.*

Proof. Let $U_1, \dots, U_n$ be a cover of B that witnesses that B is locally collared. Let $\varphi_i \colon \overline{U_i} \times [0,1] \to X$ be the corresponding embeddings. B is normal because it is compact and Hausdorff. By the shrinking lemma (already in mathlib) there is a new open cover $V_1, \dots, V_n$ of B such that $\overline{V_i} \subseteq U_i$ for all $1 \leq i \leq n$. Define

$$X^+ \coloneqq (X \amalg B \times [-1,0])/x \sim (x,0).$$

Clearly, X^+ is collared. So it suffices to show that there is a homeomorphism $G \colon X \to X^+$ s.t. $G(x) = (x,-1)$ for all $x \in B$. Let $\Phi_i \colon \overline{U_i} \times [-1,1] \to X^+$ be defined by

$$\Phi_i(x) = \begin{cases} \phi_i(x) & \text{if } x \in \overline{U_i} \times [0,1] \\ x & \text{if } x \in \overline{U_i} \times [-1,0]. \end{cases}$$

We now inductively define maps $f_i \colon B \to [0,1]$ and embeddings $g_i \colon X \to X^+$ for all $0 \leq i \leq n$ such that

(i) $f_i(x) = -1$ if $x \in \bigcup_{j \leq i} \overline{V_j}$

(ii) $g_i(x) = (x, f_i(x))$ if $x \in B$

(iii) $g_i(X) = X \cup \{(x,t) \mid t \geq f_i(x)\}$

Then g_n will be the desired homeomorphism G from above. Set f_0 to be constant 0. Then we can choose g_0 to be the inclusion and trivially $g_0(X) = X = X \cup X \times \{0\} = X \cup \{(x,t) \mid t \geq 0\}$.

Assume that g_{i-1} has been defined. Using Urysohn's Lemma, choose a continuous function $\lambda_i \colon \overline{U_i} \to [0,1]$ such that it is 0 on $\overline{U_i} \setminus U_i$ and 1 on $\overline{V_i}$. Given $x \in B$ set $a(x) \coloneqq f_{i-1}(x)$ and $b(x) \coloneqq (1 - \lambda_i(x)) f_{i-1}(x) + \lambda_i(x)(-1)$. Now let

$$s(x, \_) \colon [a(x), 1] \to [b(x), 1], \quad t \mapsto \frac{b(x) - 1}{a(x) - 1}(t - 1) + 1.$$

Now set

$$\psi_i \colon \Phi_i^{-1}(g_{i-1}(X)) \to \overline{U_i} \times [-1,1], \quad (x,t) \mapsto (x, s(x,t)).$$

This allows us to define

$$\Psi_i \colon g_{i-1}(X) \to X^+, \quad x \mapsto \begin{cases} \Phi_i(\psi_i(\Phi_i^{-1}(x))) & \text{if } x \in g_i(X) \cap \Phi_i(\overline{U_i} \times [-1,1]) \\ x & \text{otherwise} \end{cases}$$

Claim 1: Ψ_i is continuous.

Proof. It suffices to show that ψ_i is continuous. We need to show that $s \colon B \times [a(x), 1] \to [b(x), 1]$ is continuous.

Ahhh this is ugly. I somehow would prefer to just get going with this in Lean. □

□

I want to try to separate out some parts here:

Definition 1.2.8. Let X be a $(n+1)$-dimensional $\mathcal{C}^k$-manifold and $B \subseteq X$ a closed subset with some nice properties (n-dimensional). We define the topological space

$$X_B^+ \coloneqq (X \amalg B \times [-1,0])/x \sim (x,0).$$

Lemma 1.2.9. *Let X be an n-dimensional $\mathcal{C}^k$-manifold and $B \subseteq X$ a closed subset. Then X_B^+ is an n-dimensional $\mathcal{C}^k$-manifold.*

Proof. We first need to show that X_B^+ is a Hausdorff space. I am not sure if this is best done manually or if there is some cool lemma like: a pushout along closed embeddings preserves Hausdorff spaces. (Is this true?)

Then we need to show that X_B^+ is second countable. Again, there should be some general lemma about closed embeddings and pushouts.

Now we need to pick an atlas for X_B^+. Let $\mathcal{A}_X$ be a $\mathcal{C}^k$-atlas for X, $\mathcal{A}_B$ be a $\mathcal{C}^k$-atlas for B and $\mathcal{A}_{[-1,0]}$ be a $\mathcal{C}^k$-atlas for $[-1,0]$. We define

$$\begin{aligned}\mathcal{A} \coloneqq &\{(U,\varphi) \in \mathcal{A}_X \mid U \cap B = \varnothing\}\\ &\cup \{(U \times I, \varphi \times \psi) \mid (U,\varphi) \in \mathcal{A}_B, (I,\psi) \in \mathcal{A}_{[-1,0]}, 0 \notin I\}\\ &\cup \{(U \cup ((U \cap B) \times I), \varphi_I^+) \mid (U,\varphi) \in \mathcal{A}, (I,\psi) \in \mathcal{A}_{[-1,0]}, 0 \in I\}\end{aligned}$$

where we define φ_I^+ as follows. Let (U,φ) and (I,ψ) as above. Set

$$\varphi_I^+ \colon U \cup ((U \cap B) \times I) \to \mathbb{R}^n, x \mapsto \begin{cases} \varphi(x) & x \in U \\ (\varphi(y), z) & x = (y,z) \in (U \cap B) \times I \end{cases}$$

For $x \in B$ we have $\varphi(x)_I^+ = \varphi(x) = (\varphi(x), 0) = \varphi_I^+(x,0)$, so φ_I^+ is continuous.

It seems to me that this might not be $\mathcal{C}^k$ at all. I don't know of you can even glue this without basically already having the vector field available and then one might as well do the other proof. Maybe a local vector field is enough? Not sure :(

Now you would need to verify that this is a $\mathcal{C}^k$ atlas. Possibly using a version of the manifold chart lemma from Lee. I will only try to show that the transition maps are $\mathcal{C}^k$. The rest should be easier. The only interesting intersections are those of two charts of the form $(U \cup ((U \cap B) \times I), \varphi_I^+)$ and $(V, \cup((V \cap B) \times J), \psi_J^+)$. We need to check that the transition map is $\mathcal{C}^k$ on $U \cap V \cap B$. This is clear for all combinations of partial derivatives that do not include the n-th one. □

Lemma 1.2.10. *Let X be an n-dimensional $\mathcal{C}^k$-manifold and $B \subseteq X$ a closed subset. Then X_B^+ is the pushout in the category of $\mathcal{C}^k$-manifolds.*

Remark 1.2.11. It seems that this is not at all trivial and in fact much of the issue with gluing manually along boundaries. I am not sure if we can even guarantee the above lemma. The above lemma would hopefully mean that one never has to look at the explicit construction again.

Definition 1.2.12. Let X be a Hausdorff space and $B \subseteq X$ a closed subset. Let $U \subseteq B$ be an open subset together with a

We would like the following theorem:

Theorem 1.2.13. *Let M be a $\mathcal{C}^k$-manifold (with boundary) and $B \subseteq X$ a closed subset. If B is locally collared in M, then it is collared.*

Then we will need the following lemma:

Lemma 1.2.14. *Let M be an n-dimensional $\mathcal{C}^k$-manifold with boundary. Then ∂M is locally collared.*

Proof. Let $p \in \partial M$. Then p is contained in a regular half-coordinate ball U, i.e. there is a coordinate half-ball $U' \supseteq \overline{U}$ and a coordinate map $\psi \colon U' \to \mathbb{H}^n$ such that for some $r' > r > 0$ we have $\psi(U) = B_r(0) \cap \mathbb{H}^n$, $\psi(\overline{U}) = \overline{B_r(0)} \cap \mathbb{H}^n$ and $\psi(U') = B_{r'}(0) \cap \mathbb{H}^n$.

Now pick an open neighbourhood V of $0 \in \partial\mathbb{H}^n$ and $l > 0$ s.t. $\overline{V \times [0,l]} \subseteq B_r(0)$.

Considering $\overline{V}$ and V as submanifolds of $\mathbb{H}^n$ and $\psi^{-1}(\overline{V}) = \overline{\psi^{-1}(V)}$ and $\psi^{-1}(V)$ as submanifolds. Then

$$\psi|_{\overline{\psi^{-1}(V)}} : \overline{\psi^{-1}(V)} \to \overline{V}$$

and

$$\psi|_{\psi^{-1}(V)} : \psi^{-1}(V) \to V$$

are $\mathcal{C}^k$-diffeomorphisms. So it suffices to show that $\psi^{-1}|_{\overline{V} \times [0,l]} : \overline{V} \times [0,l] \to M$ fulfils the conditions given in Definition 1.2.4.

I think this should work... I think it would be nicer to do this directly in Lean and then later write up the maths proof. $\square$

which will then allow us to derive

Theorem 1.2.15 (Collar neighbourhood theorem)**.** *Let M be a C^k-manifold with boundary, then ∂M is collared.*

2 The proof of the collar neighbourhood theorem for smooth manifolds

2.1 Prerequisites

2.1.1 Vector fields

Lemma 2.1.1. *Let M be a smooth n-manifold with boundary. Let B be a coordinate half ball with corresponding map $\varphi\colon B \to \mathbb{H}^n$. Then $\frac{\partial}{\partial x^n}$ is a smooth vector field on B which is inward-pointing when restricted to $\partial M \cap B$.*

Proof. The vector field is smooth by Example 8.2 in [Lee06]. Let $p \in B$. We need to show that $\left.\frac{\partial}{\partial x^n}\right|_p$ is inward-pointing. By Proposition 5.42 in [Lee06], it suffices to show that the n-th component of $\left.\frac{\partial}{\partial x^n}\right|_p$ is positive. Its n-th component is 1 so we are done. □

Lemma 2.1.2. *Let M be a smooth manifold, A a closed and U an open subset of M with $A \subseteq U$. Let $X\colon A \to TM$ be a smooth vector field along A. Then there is a smooth vector field $\tilde{X}\colon M \to TM$ with support in U which agrees with X on A.*

Proof. We first notice that the proof of Lemma 2.26 (Extension Lemma for Smooth Functions) in [Lee06] also works for the domain TM . Now we need to verify that a smooth map of the form

General Lemma?

$$\tilde{X} = \sum_{p \in A} \tau_p(x) \tilde{X}_p(x)$$

is a vector field. Here the $\tau_p x$ are bump functions and the $\tilde{X}_p$ local extensions of the vector field. Now locally this sum is finite. Module over $C^\infty(M)$ ... blah blah blah □

Lemma 2.1.3 (Problem 8-4 in [Lee06])**.** *Let M be a smooth n-manifold with boundary. There exists a smooth vector field on M, which is inward-pointing when restricted to the boundary.*

Proof. Cover ∂M with a countable number of regular coordinate half-balls $(B_i)_i$. This is possible by Exercise 1.42 in [Lee06]. Let us call the corresponding larger coordinate half-balls B_i' and the corresponding smooth coordinate map $\varphi_i\colon B_i' \to \mathbb{H}^n$. By Lemma 2.1.1, $\frac{\partial}{\partial x^n}$ is a smooth vector field on B_i' which is inward-pointing when restricted to $\partial M \cap B$. In particular, $\frac{\partial}{\partial x^n}$ is a smooth vector field along B_i. Using Lemma 2.1.2, we can extend these to smooth vector fields $X_i\colon M \to TM$ with support in B_i' and which agree with $\frac{\partial}{\partial x^n}$ on B_i. This is again inward-pointing which is easy to see using Proposition 5.42 in [Lee06] . Now take a partition of unity subordinate to the cover $\{B_i \mid i\} \cup M \setminus \partial M$. Then do the same thing as in the proof of Lemma 2.1.2 and get a global smooth vector field that is inward-pointing on the boundary. □

write Lemma

2.1.2 Flows and Flowouts

Lemma 2.1.4. *Let U be an open subset of $\mathbb{H}^n$ that has non-zero intersection with the boundary. Let V be a smooth vector field on U. Then there exists a smooth map $\delta_U : U \to \mathbb{R}+$ and a smooth open embedding Φ_U:*

Lemma 2.1.5. *Let M be a smooth n-manifold with boundary. Let N be a smooth vector field on M that is inward-pointing when restricted to the boundary. Let (U, φ) be a boundary chart. Then there exists a smooth map $\delta_U : U \to \mathbb{R}^+$ and a smooth open embedding $\Phi_U : P_U \to M$ where $P_U =:= \{(t,p) \in \mathbb{R} \times U \mid 0 \leq t < \delta_U(p)\}$ ...*

Proof. Consider the pushforward $\varphi_* X$ of X by φ on $\varphi(U) \subseteq \mathbb{H}^n$. Extend $\varphi_* X$ to a vector field V on an open subset U' of $\mathbb{R}^n$ with $U' \cap \mathbb{H}^n = U$ such that still $V^n(p) > 0$ for all $p \in \partial\mathbb{H}^n$. Now let $\theta_V : \mathcal{D} \to U'$ be the flow of V as given by the Fundamental Theorem of Flows (Theorem 9.12 in [Lee06]) where $\mathcal{D} \subseteq \mathbb{R} \times M$ is a flow domain. Since $\varphi(U) \cap \partial\mathbb{H}^n$ is an embedded $n-1$-dimensional submanifold of U' and V is clearly nowhere tangent to it, we can apply the Flowout Theorem (Theorem 9.20 in [Lee06]) to get a smooth function $\delta'_U : \varphi(U) \cap \partial\mathbb{H}^n \to \mathbb{R}^+$ such that $\theta_V|_{\mathcal{O}} : \mathcal{O} \to U$ is a smooth embedding where $\mathcal{O} := \{(t,p) \in (\mathbb{R} \times (\varphi(U) \cap \partial\mathbb{H}^n) \cap \mathcal{D}) \mid 0 \leq < \delta'_U(p)\}$.

How? -> Lemma

show $t \geq 0$

Now define □

Lemma 2.1.6 (Boundary Flowout Theorem, Theorem 9.24 and Problem 9-11 in [Lee06])**.** *Let M be a smooth manifold with non-empty boundary. Let N be a smooth vector field on M that is inward-pointing when restricted to the boundary. Then there exists a smooth map $\delta : \partial M \to \mathbb{R}^+$ and a smooth embedding $\Phi_\delta : P_\delta \to M$ where $P_\delta := \{(t,p) \in \mathbb{R} \times \partial M \mid 0 \leq t < \delta(p)\}$ such that $\Phi_\delta(P_\delta)$ is a neighbourhood of ∂M and for every $p \in \delta M$, $t \mapsto \Phi_\delta(t,p)$ is an integral curve of N starting at p.*

Proof. Cover ∂M with a countable number of boundary charts (U, φ). □

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